Maths Exploring at Terminal 12

QCM A · transformations, amounts and averages

After S64; QCM B after S77.

For each question, only one proposal is accurate.
consult the correction.

1. $E(X) = 3$ and $Y = 2X - 1$. Then $E(Y)$ is valid for:

A. 5 B. 6

C. 2 D. 8

2. $V(X) = 4$ and $Y = -3X + 5$. Then $V(Y)$ is valid for:

A. -12 B. 17

C. 36 D. 41

3. For n independent variables of variance σ^2, the variance of their average is:

A. $n\sigma^2$ B. σ^2 / n

C. σ^2 / n^2 D. σ^2

4. Equality $E(X + Y) = E(X) + E(Y)$:

A. demands independence B. requires $X = Y$

C. is false D. applies without independence for variables finished

5. X is 0 or 2 with probabilities 1/2 each. Its variance is:

A. 0 B. 2

C. 1 D. 4

QCM B · scope of warranty

For each question, only one proposal is accurate.
consult the correction.

6. For $a > 0$, Bienaymé-Chebychev gives:

A. $P(|X - E(X)| \geq a) \leq V(X)/a^2$ B. $P(|X - E(X)| \geq a) = V(X)/a^2$

C. $P(|X - E(X)| \geq a) \geq V(X)/a^2$ D. $P(X \geq a) \leq E(X)/a^2$

7. For a Bernoulli frequency, $p = 1/2$, the bound for $P(|F_n - p| \geq 0{,}1)$ is 25 $/n$. To give it back not more than 0.05, it is sufficient to:

A. $n \geq 25$ B. $n \geq 100$

C. $n \geq 250$ D. $n \geq 500$

8. A Chebychev bound equal to 2 can only be deduced from:

A. a probability equal to 2 B. Trivial bound 1

C. impossibility D. a zero probability

9. The law of large numbers states that, for all $a > 0$ fixed:

A. the frequency becomes exactly equal to p B. Each new test reduces the error

C. $P(|F_n - p| \geq a) \to 0$ D. the observations are constant

10. If $X_1 = \cdots = X_n = X$ and $V(X) > 0$, the variance of their average is:

A. $V(X)$ B. $V(X)/n$

C. 0 D. $nV(X)$

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. A sufficient size given by Chebychev is necessarily the actual minimum size.

2. Standard deviation of $aX + b$ is $|a|\sigma(X)$.

3. Independence may be omitted in the calculation of variance of an average.